

SOFTWARE AND INFRASTRUCTURE EXPERTISE

With over 4 years of experience, I have mastered the end-to-end engineering flow: from development of software (React + Django) and deployment through DevOps practices on cloud(AWS) and on-premises(Openshift). This three-step progression reflects my expertise in building, deploying, and scaling applications seamlessly across environments of k8s clusters.

Technical Proficiency:

Development	Git , Github , Python , Django , Django-rest-framework (Rest-APIs) , Django-ORM , WSGI (Gunicorn) , Web Server (Nginx) , JWT , OAuth , Celery , Locust (stress testing) , Postman (apis testing) , HTML , CSS3 , JavaScript , react , react-redux (state-management) , axios (fetching-apis) , framer-motion (animation)
Databases	PostgreSQL , pgBackRest (backup-tool) , MySQL , Redis (in-memory db) , etcd (key-value store)
DevOps	Linux , Docker , Podman , Kubernetes (Infrastructure provisioning through Kubeadm) , Openshift Administration (without provisioning infrastructure) , Containerd , Terraform , Jenkins , CI/CD , Bash Scripting , Scripting with Python , Prometheus , Grafana
Cloud (AWS)	IAM , VPS (EC2) , EKS , VPC , S3 , RDS , Fargate (Serverless) , CloudWatch , ACM , RDS , Aurora

WORK EXPERIENCE

Role: **BACKEND DEVELOPER AND DEVOPS ENGINEER**2024/Jan-2025/Mar

Company: **KSU(King Saud University), Riyadh, Saudi Arabia**

- Designed and deployed a production-grade Kubernetes cluster using kubeadm, containerd with worker nodes across regions for fault tolerance and redundancy for python(drf) rest apis backend application.**
- Prior to deployment and infrastructure provisioning, backend application for rest apis has been developed and containerized for the orchestration tools and the Kubernetes (k8s) cluster was created using kubeadm as the core tool, containerd as the container runtime, and kubelet as the service.
- Secured k8s components with self-signed and CA-signed certificates, and established secure SSH for cluster management. And the rest apis has been proctored,secured for anonymous and application users.
- Automated CI/CD pipelines using Jenkins to extract application code from GitHub for backend application using containerize microservice for authentication, payments, scheduling, etc. with Docker, to ensure high availability.
- Provisioned infrastructure using Terraform (IaC) and deployed the multi-tier django application on AWS, ensuring zero downtime and seamless integration.
- Conducted stress testing through locust on backend django application apis created with django-rest-framework and containerized dependencies to maintain uninterrupted application performance.

Role: **SOFTWARE AND DEVOPS ENGINEER (Remote)**2022/Dec-2024/Jan

Comapny: **Direct2Success IT Solution, US**

- Provisioned AWS infrastructure using Terraform and deployed a multi-tier application on AWS EKS with a microservices architecture, integrated hybrid nodes from on-premises infrastructure.**
- Developed secure REST APIs using Python's Django-REST-Framework, integrated with Celery and Redis for asynchronous task processing and scalability.
- Designed a secure VPC with public/private subnets, leveraging AWS CloudWatch, NAT/Internet Gateways, ACM, IAM, and EC2 bastion hosts for robust and secure deployments.
- Utilized AWS EKS managed control plane and worker nodes to deploy containerized applications with restricted network policies for enhanced security.
- Configured Application Load Balancer (ALB) using AWS Load Balancer Controller (LBC) for efficient traffic routing and scalability.
- Monitored and troubleshooted Kubernetes clusters and applications using Prometheus, Grafana, and DaemonSets, deploying Prometheus within the cluster to reduce costs and centralize monitoring.

Role: **SOFTWARE DEVELOPER**2022/Mar-2022/Oct

Company: **University Of Peshawar, Peshawar**

- Developed and deployed a full-stack monolithic application on a Debian-based Linux server using JavaScript, Python, and PostgreSQL.**
- Designed a decoupled application with React frontend and Django with Django-REST-Framework backend, integrated via CORS, and deployed as individual services on Ubuntu Linux Server.
- Built dynamic UIs with React, Redux, Yup, Axios, and Recharts, and implemented secure REST APIs with JWT/OAuth (Google) authentication.
- Optimized SQL queries to ensure efficient database design and management.
- Utilized PostgreSQL database performance, managed application files and directories following Linux filesystem hierarchy (FHS) standard, and queries has been optimized inside backend django application with its ORM (Object relational mapping).

EDUCATION AND COURSEWORK

BS IN COMPUTER SCIENCE From: University Of Peshawar, Peshawar, Pakistan	2022/Sep
CKA(Certified Kubernetes Administrator) From: Udemy	2023/Jan
CKS(Certified Kubernetes Security Specialist) From: Udemy	2023/May
RHCOA(Red Hat Certified Openshift Administration (EX280)) From: OSFP	2023/Nov